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Points to Consider No. 11

Development, Classification, Manufacture, Control, and Testing of Plasmids and Vectors Used in ATMP Production

2025

PDA Parenteral Drug Association

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Introduction

Cell and gene therapies are classes of advanced therapy medicinal products (ATMP) that include numerous different types of products including nucleic acids, viruses, genetically modified microorganisms, and others. Gene therapies are relatively novel technologies that tend to have extremely complex and not well-understood mechanisms of action but hold great potential for treating a variety of serious diseases, unmet medical needs, or simplification of administration and treatment (i.e., "one shot" gene therapy treatments versus a lifetime of chronic treatment for many diseases). The potential applications are diverse and numerous and may reach far beyond any current forms of therapies in terms of disease-state targets, as illustrated by the impact of the handful of currently approved successes. Notable, as well, are the abundant early- to late-stage products currently in clinical development and the anticipation of many more to come.

One technology that has shown immense promise within the ATMP space is the use of viral vectors for delivery of a therapeutic gene. Viral vectors have been shown to be useful both in the direct administration of in vivo treatment to patients and in the ex vivo genetic modification of cells that are then administered to the patient. Different viral-vector types have proven effective as gene delivery systems including, but not limited to, adeno-associated viruses (AAV) and lentiviruses, both of which have successfully commercialized products. AAV-based drug products, in particular, have been commercialized successfully in the United States, Europe, and other regions of the world.

Plasmid DNA (pDNA) are used as critical materials (i.e., the "building blocks") in the production of viral vectors and may also be delivered as naked DNA or encapsulated (e.g., in a liposome) for delivery purposes at various steps of a gene-therapy manufacturing process. Thus, ensuring consistent control of the quality in the production and use of plasmids is of great importance in the manufacture of many gene therapies, including cell-based, gene-modified therapies.

The ex vivo modification of cells was not within the primary scope of *PDA Technical Report No. 81: Cell-Based Therapy Control Strategy* (TR 81), therefore, some points to consider on viral vectors used in gene therapy were deemed to be of value and are addressed herein. In developing this points to consider document, the intent was to not repeat information already available in TR 81, but rather to supplement it. Likewise, when topics such as process validation, sterile filtration, filtration for viral contaminants, analytical method validation, and phase-appropriate GMPs are addressed in this document, these topics should be deemed supplemental to guidance that can be found in existing PDA technical reports.

A team of industry representatives from multiple CMC disciplines comprising product and process development, analytical development, manufacturing, quality assurance, and regulatory affairs was assembled by the PDA ATMP Advisory Board to author this document. This team was tasked not only with identifying the current major problems in the development, qualification, and manufacture of viral vectors that are not adequately addressed in current regulatory guidance, but also with recommending solutions to

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these problems based on current industry perspectives and best practices with consideration to compliance within a dynamic, global regulatory framework.

This document presents the views of the Parenteral Drug Association on the development, classification, manufacture, control, and testing of plasmids and vectors used in ATMP production. It is designed to encourage further dialog with industry, health authorities, and suppliers of technology and materials while taking into consideration the changes and needs of the modern, global, sterile healthcare-product manufacturing industry. It does not represent a standard or a regulatory guidance.

Scope

This Points to Consider document is focused on the use of viral vectors for the purpose of the delivery of a therapeutic gene both in vivo and for ex vivo modification within the overall classification of ATMPs. The topics addressed are diverse and include plasmid and viral vector categorization and control, quality by design, control strategies, filtration, comparability, platform technologies, and potency assays. Separately and together, these topics address a broad range of development, manufacture, and control issues that may be encountered in industry. The topics are organized in a question-and-answer format, presenting a specific problem statement and an accompanying recommendation, inclusive of the recommended points to be considered with the goal to assist the reader in formulating a strategy to address the topic.

One concept that emerges throughout these discussions is the need for applying a science-driven and risk-based approach to providing solutions for current problems associated with the development and manufacture of viral vectors. That approach is embraced by regulatory agencies for the development of gene therapies, including those using viral vectors, as presented in the EU *GMP Annex 1: Manufacture of Sterile Medicinal Products and the Guideline on the Risk-based Approach according to Annex I, Part IV of Directive 2001/83/EC* ***(1)***. Annex I, Part IV specifically refers to a risk-based approach as the recommended methodology to design the organizational, technical, and structural measures to comply with GMP regulations. PIC/S *Annex 2A: Manufacture of Advanced Therapy Medicinal Products for Human Use* echoes the EU guidelines, noting that quality risk management principles are particularly important for this class of materials ***(2)***.

In summary, the topics presented in this Points to Consider reflect the current thinking regarding some of the major challenges that have been identified by the PDA community regarding the development, control, and manufacture of gene therapy products using viral vectors. Nonetheless, this team acknowledges that there are many more topics that the industry faces for this diverse class of therapies.

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Topic 1. Categorization of Viral Vectors across Regulatory Jurisdictions and their Control Strategies

Q 1.1. Is there a framework for classifying plasmids and viral vectors such as starting materials, critical starting materials, or drug substances?

Plasmid and viral vectors (i.e., vectors that consist of or are derived from biological material designed to deliver genetic material) express the desired gene product that is introduced into the ATMP manufacturing process at different stages of production. In some cases, the purified genetic material (i.e., plasmid or vector) is administered directly to a patient whereas in other cases, the genetic material, most commonly a viral vector, is used to genetically modify the cells ex vivo as part of the cell-therapy manufacturing process. This leads to a situation wherein similar material can be classified as a raw material, starting material, critical starting material, or an active substance (drug substance or active pharmaceutical ingredient) with each governed by different GMP requirements within different regulatory frameworks. The regional regulatory jurisdiction(s) will dictate the classification requirement, and therefore, the governing GMPs.

A starting material may be deemed a critical starting material depending on an assessment of risk and the particular regulatory jurisdiction. A critical classification may drive tighter controls within the quality system. For example, in the European Economic Area, a viral vector that is used to modify the cells ex vivo will be considered a critical starting material and follow EU *EudraLex - Volume 4 - Good Manufacturing Practice (GMP) Guidelines*, while in the United States, it may be considered a drug substance ***(3)***. In some jurisdictions, such as Australia, the difference is even more significant: if the vector is administered in vivo to the patient, it will be considered a medicine and follow the PIC/S code of GMP PE-009 (which is largely aligned with EU Volume IV with part 4 being a notable exception) ***(4)***. However, the if same viral vector is used to modify the cells ex vivo, it will be considered a critical starting material and is regulated by a completely different code of GMPs (refer to the Australian *Code of Good Manufacturing Practice for Human Blood and Blood Components, Human Tissues and Human Cellular Therapy Products*) ***(5)***. Considering that various regulatory agencies have different GMP and CMC requirements, it is extremely important to be aware of these nuances to determine the correct classification for each material used during manufacturing in order to develop the appropriate control strategy.

While it is not possible to provide a comprehensive framework for categorizing vectors and plasmids, as there are too many jurisdictions with varying requirements, it is possible to show approaches that would facilitate classification of the material, establish the CMC and GMP requirements, and build an appropriate control strategy. That approach can then be adapted for other jurisdictions.

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Q 1.2. What guidance is available for the classification of plasmids and viral vectors?

Some guidance for classifying vectors and plasmids is available from U.S. and European regulators, respectively. In particular, the first two guidelines listed below can be considered primary references, though the FDA guidance refers to investigational products while the EMA guidance focuses on approved products. While neither guidance provides a clear indication that the principles outlined can be applied in a phase-appropriate manner, industry consensus and informal regulatory authority commentary suggest that a phase-appropriate strategy should be implemented.

- U.S. FDA Guidance for Industry: Chemistry, Manufacturing, and Control (CMC) Information for Human Gene Therapy Investigational New Drug Applications (INDs)
- U.S. FDA Guidance for Industry: Considerations for the Development of Chimeric Antigen Receptor (CAR) T Cell Products
- U.S. FDA Guidance for Industry: Human Gene Therapy Products Incorporating Human Genome Editing
- European Medicines Agency (EMA) Guideline on the Quality, Non-Clinical and Clinical Aspects of Gene Therapy Medicinal Products (EMA/CAT/80183/2014)
- EMA Guideline on Quality, Non-Clinical and Clinical Aspects of Medicinal Products Containing Genetically Modified Cells (EMA/CAT/GTWP/671639/2008)
- EMA Questions and Answers on the Principles of GMP for the Manufacturing of Starting Materials of Biological Origin used to Transfer Genetic Material for the Manufacturing of ATMPs (EMA/246400/2021)[1]

[1] The EMA Q&A is based on the PIC/S Annex 2A revision with a minor difference: PIC/S Annex 2A (Sections 5.20-5.29) includes some clauses that can be used systematically to classify the material. Those principles are mandated only where PIC/S Annex 2A is approved, but the authors believe these principles can be scientifically applied to build a strategy that can be defended for multiple jurisdictions.

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Q 1.3. What are the principles of plasmid and viral-vector classification summarized in the regulatory guidance?

While principles from the regulatory documents identified in **Q 1.2** are summarized here, each regulatory body offers subtle differences on various points and sponsors should carefully check the latest guidance available in the applicable jurisdiction(s). When targeting multiple jurisdictions, the sponsor should consider the most restrictive regulatory environment where they intend to market their product. This approach should also consider that jurisdictions with less restrictive requirements may still have jurisdiction-specific requirements that would also need to be fulfilled. For example, when manufacturing a critical starting material, establishing a master virus seed stock or master cell bank will enable the sponsor to enter each jurisdiction without delaying the program to remanufacture the critical starting material. In fact, in Europe, having a master stock/bank is becoming required, even at an early stage. Therefore, it would be advantageous for the firm to rapidly progress to reference stocks/banks with an eye towards manufacturing the master stock/bank as quickly as possible.

Figure 1.3-1 presents the three types of gene therapy drug products:

1. In vivo gene therapy (e.g., mRNA or DNA)
2. In vivo gene therapy using viral vectors (e.g., AAV)
3. Gene therapy where a cell therapy product is modified ex vivo (e.g., lentiviral vector genetic modification of T-cells to make a chimeric antigen receptor T-cell therapy)

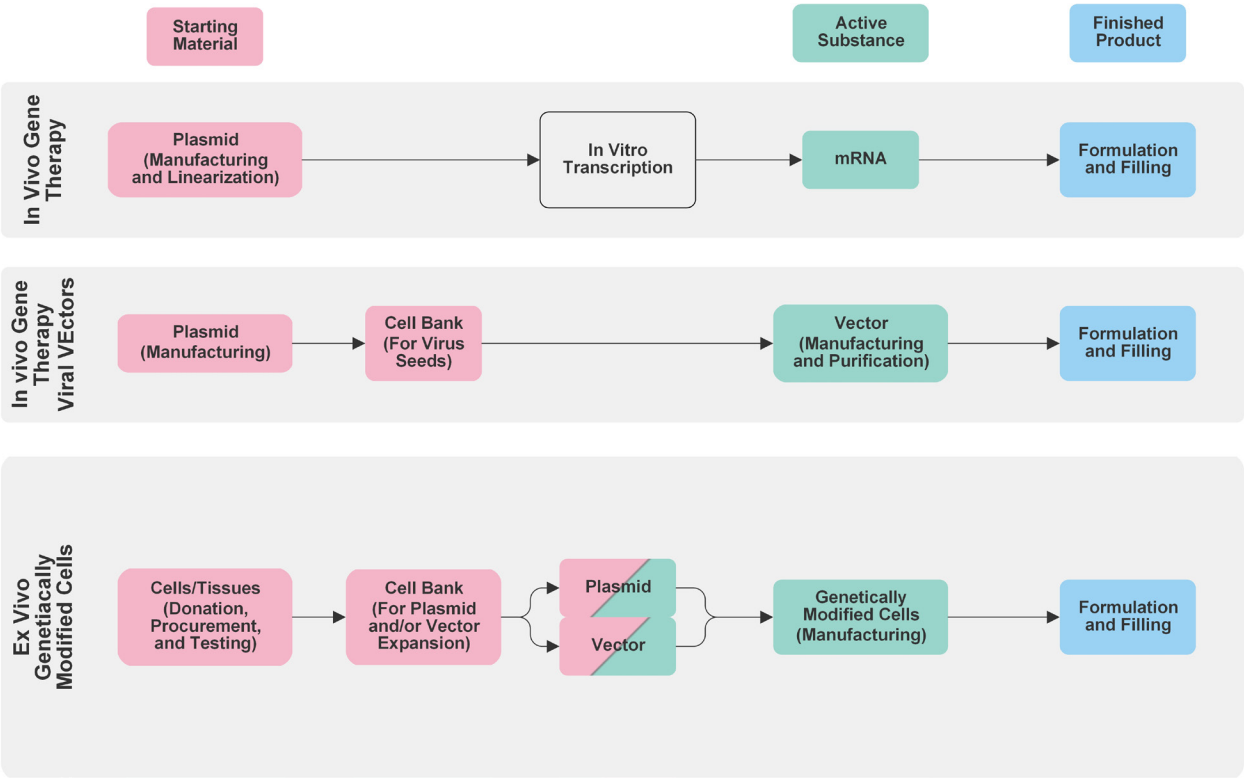

Figure 1-3.1 Graphical representation of starting materials (shown in pink), drug substances (shown in green), and finished product (shown in blue)

For mRNA- or DNA-based in vivo gene therapies, which are directly administered to patients, the mRNA or DNA used for injection is the active drug substance(s), and the formulated, filled product in the primary

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container is the drug product. In these situations, the starting materials include the plasmid used at the start of the drug-substance manufacturing process and the bacterial master or working cell bank that is used to expand the plasmid prior to routine drug-substance manufacturing.

Similarly, for an in vivo viral-vector gene therapy, the plasmid used to transfect production cells or to make the master virus seed stock and master cell bank for vector production is the starting material, and the actual viral-vector produced is the active substance.

Finally, for ex vivo genetically modified cells, the FDA considers both the viral vector and the genetically modified cells as active drug substances, so the sponsor must file two drug substance sections. The EMA, however, considers genetically modified cells the only active substance requiring a drug substance section. Intriguingly, the EMA Q&A document specifies that the vector is a starting material for which the principles of GMP for starting materials are applicable ***(6)***. Question 2 in the EMA document asks: "What does principles of GMP mean?" which is answered as follows:

> GMP requirements for ATMP active substances (drug substances) and ATMP finished products are described in Part IV of the GMP Guide. Some requirements of Part IV can also be relevant for the starting materials.
>
> However, the main differences with regard to the starting materials are:
>
> - Not all GMP aspects described in Part IV of the GMP Guide are required.
> - According to the current legal framework, neither recurring inspections nor GMP certifications by the responsible authorities are required.
> - As a result of this situation, the ATMP manufacturers have the responsibility to verify that appropriate GMP requirements are implemented for the manufacturing/testing of the starting materials according to the methodology described in questions 4 and 5 of the document.

The EMA document asserts that the sponsor is accountable to apply a risk-based analysis to identify which parts of Part IV of the GMP Guide are applicable, and that the sponsor's analysis, supplier qualification, and starting-material quality are assessed during clinical trial application, marketing authorization, variation application, and GMP inspections of the ATMP manufacturer.

Aside from the nomenclature differences between the FDA and EMA guidances, for any sponsor considering clinical trials in both the United States and European Union, the viral vector should be treated like an active substance that is manufactured according to GMPs and should be filed as a separate drug substance section within the IND application for FDA. This standard will meet the EMA requirements and thus provide access to both marketplaces as needed.

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Q 1.4. What are the GMP and regulatory requirements for vectors defined as active drug substances and starting materials?

The FDA CMC guidance states that:

> You should describe whether the vector DS will be formulated into the DP for administration of the genetic material (section V.B.3.b., "Batch Formula (3.2.P.3.2)," of this guidance) or whether it will be formulated as a bulk DS for ex vivo genetic modification of cells (as outlined in section IV.B., "Drug Substance and Drug Product," of this guidance). As an active pharmaceutical ingredient, an appropriate level of control should be applied to each DS, and each DS should be manufactured under current good manufacturing practice (CGMP) conditions that are appropriate for the stage of development (Ref. 11) ***(7)***.

Therefore, clearly the FDA considers the vector an active pharmaceutical ingredient, equivalent to a drug substance, for which phase-appropriate GMPs are required.

The EMA provides a more explicit list of factors to include in the risk-based analysis of the starting-material manufacturing which may be useful to consider for this question. These considerations apply to any vector manufacturing process, whether performed by the sponsor or by a contract manufacturing organization. Question 5 of the EMA Q&A directs that consideration should be given to ***(6)***:

- Cross-contamination with other vectors or genetic material
- Adventitious agents, for example:
  - Transmissible spongiform encephalopathy such as Creutzfeldt-Jakob disease from human donors, or bovine spongiform encephalopathy from bovine serum raw materials
  - Viral contamination or recombination into replication competent virus
- Microbiological controls for sterility, mycoplasma, or endotoxins
- Impurities:
  - Originating from raw materials added to the process (e.g., antibiotics)
  - Generated as part of the process (e.g., addition of transfection enhancers)
  - Carried over into the process (e.g., residual plasmids that are carried over from the vector production into the cell-therapy process)
- Environmental controls
- Storage and transportation conditions, including supply-chain complexity
- Stability

The overall process must be taken into consideration when evaluating the starting materials for any active drug substance that is similar in size to, or larger than, a small virus. This is equally true for a directly administered gene therapy as it is for a viral vector used in ex-vivo genetic modification.

For these active substances, removing all risk from adventitious viruses (e.g., parvovirus) from the manufacturing process is not possible and therefore, the adventitious-agent control strategy for the starting material is paramount. While both the FDA and EMA accept animal- and human-derived raw materials (e.g., bovine or human albumin serum), they should be avoided where possible or be gamma-irradiated, unless that process impacts the product quality.

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In another example, human-derived raw materials used to manufacture starting materials may have different donor-screening requirements depending on jurisdiction. Sponsors and the ATMP manufacturer should pay special attention to the donor location, screening, testing, and pool size, as well as viral removal and inactivation steps during processing. For example, additional testing may be required for the raw material itself or a negative test may be required from the donor. The control strategy should again address the most stringent requirements from the intended jurisdictions.

As stated in the FDA CMC guidance, "As the sponsor of the IND, you are ultimately responsible for the safety of subjects in the clinical investigation (21 CFR 312.3); therefore, we recommend that you and the [Contract Manufacturing Organization] CMO understand and document your respective responsibilities for ensuring product quality" ***(7)***.

To ensure the safety of subjects, it is imperative that the sponsor receive all necessary information and documentation from the CMO required to complete the IND sections relevant to the manufacturing process to properly assess and control product quality and safety. The responsibilities for this knowledge transfer should be delineated within the context of a Quality Agreement to ensure that each organization involved in manufacturing is aware of their activities and how these activities will be conducted in compliance with GMPs. The 2016 FDA *Guidance for Industry: Contract Manufacturing Arrangements for Drugs: Quality Agreements* provides a framework for establishing a Quality Agreement and includes examples of how quality management principles can be used to aid in the management of a contract manufacturing relationship ***(8)***.

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Q 1.5. What are the special considerations for the classifications of plasmids?

Plasmids are critical raw materials for most gene-therapy-based ATMPs, as shown in **Figure 1.3-1**. It is imperative that the plasmid contains the desired transgene and associated genetic elements (e.g., promoters) which have the correct nucleotide sequence since any mutation, deletion, or insertion could significantly impact the expressed protein sequence, function, or expression level. This is true for both DNA-containing therapies and mRNA therapies, where the plasmid is reverse-transcribed into the mRNA.

Assessing the classification of plasmids is also important in the context of viral vectors (either for direct injection as a gene therapy or for ex-vivo genetic modification of cells) as the viral-vector manufacturing process may use three or more plasmids to mitigate the risk of recombination that could form a replication-competent virus. As examples, AAV- and lentivirus-based vectors have a common manufacturing approach that entails the co-transfection of host cells using multiple plasmids to separately encode the gene of interest, the viral structural proteins, and the viral helper genes required to assemble the viral vectors. As a result of the multiple plasmids, there are multiple sources of error or contamination that could impact patient safety and product efficacy.

Many manufacturers outsource plasmid manufacturing to contract development and manufacturing organizations (CDMOs) which, in turn, manufacture plasmids for different clients with different virus types and sub-types. Potential cross-contamination could occur due to shared equipment and materials, personnel flow, or insufficient cleaning during changeover. Indeed, clinical trials have been put on hold due to contamination of research-grade plasmid used in producing clinical materials. The sponsor needs to ensure that risks are assessed, preventive actions are taken to avoid such potential cross-contaminations, and that appropriate methods are established to confirm and characterize the identity and quality of the plasmids being used in the sponsor's manufacturing process.

The FDA *Guidance for Industry: Human Gene Therapy for Neurodegenerative Diseases* includes the expectation that plasmids manufactured in a multiproduct manufacturing facility should be tested for the presence of other contaminating plasmids which may have been co-purified ***(9)***. Implementation of such testing may pose logistical challenges to the sponsors as CDMOs typically do not share information regarding other clients' production schedules, let alone proprietary information (e.g., plasmid sequence) that might be needed to detect contamination. As part of the Quality Agreement expectations, sponsors should require their CDMOs to demonstrate robust changeover procedures from product to product, validate cleaning and sanitization processes, and ensure other procedural controls are in place to prevent cross-contamination. Identity and/or purity assays should include criteria that can detect the occurrence of cross-contamination. Sponsors may also consider sequencing the CDMO product periodically, whether a plasmid or viral vector, as part of raw-material receipt by the sponsor's ATMP drug product manufacturer to support a holistic control strategy established to ensure product identity.

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Topic 2. Quality-by-Design (QbD) Principles

Q 2.1. What is quality-by-design (QbD) and how does it apply to viral vectors and plasmids?

Quality-by-design (QbD), first developed by Dr. Joseph M. Juran, has evolved greatly since its inception and today it is supported by such widely recognized standards as the International Council for Harmonisation's Quality Guidelines *ICH Q8(R2): Pharmaceutical Development*, *ICH Q9: Quality Risk Management*, and *ICH Q10: Pharmaceutical Quality System* ***(10-12)***. The guiding principle of QbD asserts that quality should be designed into a product and into its manufacturing process. By designing the product and process properly, understanding is increased, thereby simplifying the troubleshooting of process and product problems in the future. This perspective is encouraged by the FDA and by other global regulators. This enhanced process knowledge, in turn, simplifies risk-based quality decision-making, as the risk-based decisions are made within the framework of the effective boundaries of the process.

The major deliverables of QbD include:

- Quality target product profile (QTPP) that identifies the critical quality attributes (CQA)
- Product design that includes identification of critical material attributes (CMA)
- Process design that includes identification of critical process parameters (CPP)
- Control strategy that links CPPs and CMAs to the delivery of CQAs
- Process capability
- Continuous quality improvement

Like large-molecule biotechnology products and small-molecule conventional therapies, viral vectors and plasmids benefit from a structured QbD approach to development and continuous improvement. Product development teams make use of prior platform experience and scientific knowledge (e.g., industry standards, published examples) as well as perform studies as needed (e.g., design of experiment) to create the body of knowledge (subject to rugged data analysis) which supports the product throughout its lifecycle. That data is then subject to knowledge management principles and is updated and enhanced by production experience and stability results to support commercialization. At the end of each phase of pharmaceutical development, stage gates should be defined to encourage revision of the QTPP, control strategy, and control ranges. Final revision of the overall initial control strategy should be performed at the conclusion of the process performance qualification. Lifecycle maintenance of the control strategy will then be required until product retirement.

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Q 2.2. What aspects of quality-by-design (QbD) are the marketing authorization holder's responsibility when working with a contract research organization (CRO) or contract manufacturing organization (CMO)?

The determination of responsibility for QbD is commonly part of the business agreement and Quality Agreement between the (future) registration holder and the contract research organization (CRO) or contract manufacturing organization (CMO). However, the latitude to shift responsibility from the marketing authorization holder (MAH) or the sponsor to the CRO or CMO is limited as some responsibilities cannot be delegated and must stay with the MAH or sponsor (e.g., disposition decisions). The EMA *Reflection Paper on Good Manufacturing Practice and Marketing Authorisation Holders* can help in understanding the guiding principles ***(13)***. This reference applies to the European Economic Area (EEA), and compliance with it has brought demonstrated success with other jurisdictions.

When making use of a CRO for the development of the product, the QTPP should be prepared by the Sponsor (i.e., future MAH) and reviewed with the CRO. Since the QTPP represents the development target from a quality perspective, alignment and mutual understanding on its content are essential. Typically, it is then the responsibility of the CRO to conduct the necessary studies to determine the preliminary CQAs, CMAs, and CPPs. The (future) MAH will be required to periodically assess and verify that the recordkeeping associated with the development activities are sufficient to support the development, technology transfer, product registration, and future product disposition. Ideally, the Quality Agreement should specify how the documentation is developed, recorded, and archived as appropriate to the phase of development. In particular, a written agreement should be struck that identifies the work (both manufacturing and analytics) to be performed under the GxP principles pertaining to the appropriate phase of development/clinical-batch production. It is critical to ensure that there is an explicit agreement about who owns the data and important to define that the Sponsor retains access to the intellectual property (IP) associated with the developed methods or processes. Without an explicit agreement, a risk exists that a successful product may become legally difficult to progress. (For more information, see PDA *Technical Report No. 56 (Revised 2016): Application of Phase-Appropriate Quality System and cGMP to the Development of Therapeutic Protein Drug Substance (API or Biological Active Substance)*).

When the (future) MAH has concluded product development and transfers the product to a CMO[2] for commercialization, the sending unit (MAH) must provide the QTPP, CQAs, CMAs, and CPPs established during development to the receiving unit (CMO). These specifications and parameters will be codified in the technology transfer protocol(s) and then will be introduced into the GMP documents for the CMO (e.g., specifications, master batch records, analytical methods, and analytical method and process validation protocols). Additional updates to the control strategy may be made as the specifics of the equipment, automation, and controls are explored during transfer and subsequent validation. Document control processes will ensure that appropriate change control and version control are maintained for the controlled documents that define the specifications and control strategy. Comparability will be required for any new site or significant change in product or process. (See **Topic 5.**) Process capability will then be established with the technology transfer and process validation, becoming the initial benchmark for future continued process validation and annual product review. (For more information, see PDA *Technical Report No. 65 (Revised 2022): Technology Transfer*.)

[2] When the CRO and CMO are the same entity, technology transfer should still be a formal stage-gate for development and the essential documentation and controls should be verified to exist before the production of batches.

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Q 2.3. What aspects of QbD apply if the firm sources viral vectors or plasmids rather than producing them?

Before determining which aspects of QbD apply, it is important to understand whether the material will be considered a starting material or an active substance (e.g., drug substance), and the jurisdiction in which it will be regulated. If regulated as an active substance, in specific jurisdictions, the requirements of a CMO as described in **Q 2.2** may apply.

However, if regulated as a critical starting material, the requirements may differ. Critical starting materials are required to have an incoming specification and suitable test methods to determine, among other things, the identity, purity, and whatever attributes of potency or functionality can be assessed at this stage in production. There may be requirements for viral and microbiological safety of the material as well. However, risk assessments may be appropriate to evaluate the criticality of the starting material and whether all these requirements should apply. As with all supplied materials, it may be possible to receive some or all of these test results based on the supplier's Certificate of Analysis (CoA) and Certificate of Conformance, provided the supplier has been qualified (and/or audited) specific to their manufacture, control, and testing and subjected to an appropriate risk assessment. Identity-testing, however, is a phase-appropriate GMP requirement and the nature and extent of the identity-testing performed should be defined and justified. For example, in some cases, partial sequencing may provide sufficient discrimination of identity without having to perform testing of the full-length sequence.

In cases where the manufacturing platform or active substance has not been developed on behalf of the (future) MAH for a specific clinical trial or application, the supplier (i.e., the CMO who supplies the viral vector or plasmid) may hold the intellectual property rights. Global regulatory health authorities may not have a final standardized guidance to address how such a product will be filed and reviewed: therefore, it is important to actively communicate with the CMO and the regulatory health authority in question. In some cases, the supplier may not be willing to provide the same level of detail about the process to the (future) MAH as a true CMO, which may be appropriate given the phase of the product's development. In these cases, the basic details should be made available by the supplier to support a clinical trial dossier. In other cases, regulators may communicate that those details of a manufacturing platform process should be shared to support a marketing authorization. This may depend on how the material is classified and the jurisdiction pertaining to the application. In any case, onsite auditing will be required to inspect the original source data that supports those technical details.

Viral vectors and plasmids are considered critical to the manufacturing process. Even in a jurisdiction that regulates these as starting materials, compliance to the manufacturing process and manufacturing conditions of the supplier are essential quality elements. Therefore, the quality management program of the (future) MAH should appropriately classify the supplier as critical and require a stringent onsite audit to ensure material quality and adherence to GMPs in a phase-appropriate manner. In addition, due to the assigned criticality, some jurisdictions may reserve the power to conduct risk-based inspections of the facility to ensure GMPs are followed. It is advantageous for these materials to be "banked," which means that characterization of the material, stability during storage, and effective inventory management principles to reduce the risk of mix-up are all essential elements of the supplier- and self-audits. In addition, qualification of the analytical methods and equipment that established the master/working bank(s) must be completed. All these controls should be implemented in a phase-appropriate manner. For more information, see PDA *Technical Report No. 56 (Revised 2016) Application of Phase-Appropriate Quality System and cGMP to the Development of Therapeutic Protein Drug Substance (API or Biological Active Substance)*.

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Whether the viral vector or the plasmid is considered an active substance or a starting material, change control management must be established by the supplier and, as part of the Quality Agreement, details of product, process, or analytical changes must be made known to the MAH or Sponsor prior to change implementation so any impact to the registration or clinical trial application (or its equivalent) may be assessed. When the product, process, or analytics are not the intellectual property of the MAH or Sponsor, sufficient details should be provided to appropriately assess impact and to inform regulators when deemed appropriate.

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Topic 3. Control Strategies

Q 3.1. Are control strategies required for viral vectors and plasmid manufacture? If so, at what stage should the control strategy be available?

Viral vector and plasmid manufacture, whether defined as a critical starting material or as a drug substance, requires a robust, holistic control strategy to drive product quality. The control strategy should link the CPPs and CMAs directly to the CQAs. By understanding which process parameters have a direct impact on the CQAs, appropriate process monitoring and controls can ensure reproducible and predictable delivery of the viral vector or plasmid. PDA *Technical Report No. 81 Cell-Based Therapy Control Strategy* (TR 81) describes the approach for cell-based therapy control strategies; the strategies and concepts described in PDA *Technical Report No. 60 Process Validation: A Lifecycle Approach* (TR 60) apply to all pharmaceutical products. The principles discussed in TR 81 and TR 60 can broadly be applied to viral vector and plasmid control strategies as well.

In addition to the principles mentioned above, phase-appropriate implementation may be warranted. For example, the control strategy may consist of preliminary factors to be monitored and assessed in a phase-appropriate manner. As experience with the production of Phase 1 and Phase 2 material is gained, the control strategy may mature while regional differences may dictate how early the control strategy must be formalized. As additional product and process knowledge is gained through clinical results and through continued small-scale and lifetime studies, process ranges must be established and CPPs defined, so that by the time pivotal clinical batches are produced, product and process consistency are assured. When creating a holistic control strategy for plasmid and vector production, inclusion of microbiological and viral control aspects, in addition to the core manufacturing process elements, is important. The microbiological and viral control strategies can either be prepared as separate documents or they may be written within the overall process control strategy. Either approach is acceptable, provided that all stages of the process and all material handling steps are included in the approaches.

Viral and microbiological controls should be in place for the first human clinical trial as they have a direct impact on patient safety. Therefore, these attributes of the control strategy may need to mature earlier than some of the other parameter definitions. In the EU, sterilization and sterility methods are required to be validated at the time of Phase 1 for an Investigational Medicinal Product Dossier. For example, if a vector is defined as a sterile critical starting material, safety-indicating methods (such as sterility and endotoxin) must be qualified. Requirements for a mycoplasma safety-indicating method may vary based on regional requirements. A risk assessment may be needed which takes into consideration the intended market and existing controls. Additionally, the initial control strategy will need to be refined and revised as additional product knowledge and process understanding are gained.

Analytical methods and their associated reference standards should mature along with the control strategy. As CPPs are defined for each associated CQA, the analytical methods that confirm CQA achievement must

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be demonstrated to be reproducible, accurate, and precise. In this manner, should late-stage process changes or process parameter changes be introduced (within or between sites), comparability and bridging can be accomplished with a high degree of assurance. (See **Topic 5.**)

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Q 3.2. What aspects of microbiological control apply to vector and plasmid manufacture?

Phase-appropriate microbiological- and viral-control principles should be applied regardless of whether the product is considered a starting material, active substance, or final product. During the definition of the QTPP, it should be established whether the viral vector or plasmid is intended to be sterile or low bioburden. This definition will guide which aspects of microbial control to apply to the process.

After classification as sterile or low bioburden, the microbiological control strategy should then be established based on a systematic assessment of each process step. The assessment should consider the intrinsic and extrinsic sources of contamination with a focus on any open processing. The control strategy must identify appropriate active controls (e.g., incoming testing or in-process treatment, such as filtration) to limit microbiological and viral contamination. In addition, monitoring requirements (e.g., environmental monitoring, bioburden sampling) should be defined and implemented as passive measures to ensure that the state of control is maintained. The intent is to establish both active and passive measures that provide confidence in the assurance of the claim of either sterility or low bioburden.

Aseptic controls are critical requirements for operation. For low-bioburden processes, control strategy development can focus on which aseptic controls are essential to support a compliant low-bioburden material. Open processing and manual manipulation steps should be a particular focus in the microbiological control strategy. Certain factors in functionally closed processes, as described in EU Annex 1, should also be considered.

Whether the end product is considered sterile or low bioburden, all basic elements of a contamination control strategy (CCS) should exist, including:

- Environmental monitoring program (see PDA *Technical Report No. 13 (Revised 2022) Fundamentals of an Environmental Monitoring Program*)
- Cleaning and sanitization of the environment (see PDA *Technical Report No. 70 Fundamentals of Cleaning and Disinfection Programs for Aseptic Manufacturing Facilities*)
- Specified personnel and material flow that prevent cross-contamination
- Gowning
- Access control
- Environmental controls (e.g., temperature/humidity monitoring, pressure cascades, routine certification of HEPA filters)
- Control of raw materials
- In-process and final-product testing
- Cross-contamination controls, including viral material

Additional elements of the CCS should be risk-based and specific to the process and the end claim. Examples of differences in control strategy elements, based on a product sterility claim, are presented in **Table 3.2-1**.

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Table 3.2-1. Examples of Control Strategy Elements Based on Product Classification

Low-Bioburden	Sterile/Aseptic
Extensive initial qualification of the microbial control of the facility with ongoing environmental monitoring of the process	Aseptic process simulation/aseptic process validation initially and semi-annually for each process
Risk-based screening of incoming material for viral and microbiological contaminants and/or acceptance of vendor CoA	Screening of incoming materials for virus and microbiological contaminants
Less-stringent gowning requirements	In-process bioburden sampling
Routine environmental monitoring to demonstrate room classification	Aseptic gowning and gowning qualifications
Bioburden and endotoxin testing of the final product	Continuous environmental monitoring during open processing for each batch
	Open processing steps to be performed in a Grade A/ISO 5 environment, along with the controls required to maintain this environment
	Sterility and endotoxin testing of final drug substance and/or drug product

Within a given control strategy, there may be different approaches to achieve the same level of quality. For example, some firms include screening patient materials for bioburden and endotoxin in their control strategy while other firms set aside retention samples to be used in the case of a manufacturing investigation. The utility of the latter approach is that it speeds any possible investigation without consuming day-to-day resources for low-risk, single-use equipment with closed processing. The former approach demonstrates an active surveillance program to detect potential contamination which could be introduced to the facility.

In addition to supporting multiple approaches to the same issue, control strategies may mature or change as development progresses or as experience is gained. For example, during clinical-trial eligibility screening, patients are tested for multiple viruses, serving as a beneficial control. Once commercial production begins, some facilities consider adding adventitious-virus testing, thereby enhancing the surveillance and control. In both cases, the regulatory requirements form the minimum basis of the control strategy.

Many viral vectors and plasmids cannot withstand terminal sterilization or sterile filtration, which puts them in a high-risk category due to the reliance on the aseptic processing to safeguard the sterility of the final product. For sterile processes, aseptic process simulation (APS)/aseptic process validation (APV) is expected to demonstrate appropriate control over aseptic operations. This is the case for both drug substance and drug product, although the power of the study may be less with drug substance. In some jurisdictions, such as the United States, the expectation at Phase I is that the control strategy demonstrates the capability

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of the aseptic process but does not strictly require a simulation run with a challenge fluid. As the product progresses to Phase II and beyond, the first full validation will be conducted. For other jurisdictions, such as the EU, it is expected in Phase I that the process is fully validated at this early time-point. Whether water trial or media simulation, the expectation is that the process simulation is risk-based, with an emphasis on all open and manual manipulations as well as sterile holds. Where unit operations or manufacturing steps are considered "functionally closed" (e.g., steps that are closed during processing but not fully closed during support activities such as sampling), simulating these steps is recommended so data can be generated that demonstrates the necessary level of control over the process. As with all aseptic simulation activities, it should be remembered that it is not possible to validate poor aseptic technique, and that the methodology is probabilistic. The APV/APS provides confidence that the process is capable, while the routine monitoring provides the assurance of the day-to-day controls.

See *PDA Technical Report No. 62 Recommended Practices for Manual Aseptic Processes* for additional details on APS and APV as they apply to manual aseptic processing. See *PDA Technical Report No. 90 Contamination Control Strategy Development in Pharmaceutical Manufacturing* for additional details on development of a CCS.

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Topic 4. Filtration

Q 4.1. Should filtration be considered as a part of the initial strategy of development and control?

For contamination control of proteins and biologically active materials, the use of filtration technologies is commonly preferred over other technologies (e.g., steam sterilization, gamma sterilization, X-ray) because it tends to reduce the impact on the product's quality attributes.

Microbial and viral filtration should be included in the design of the manufacturing process from the earliest conceivable timepoint so that its introduction later in development does not introduce any unintended impact on product quality attributes. Establishing filtration for a process may be challenging, as lentiviral, rHSV, and gamma-retroviral vectors may be damaged by filtration or result in such low potency that a lot may be unusable. While AAV vectors can typically be filter-sterilized, potential product changes due to filtration, such as aggregation, should be characterized.

PDA *Technical Report No. 26 Sterilizing Filtration of Liquids* (TR 26) and *Technical Report No. 41 (Revised 2022) Virus Retentive Filtration* (TR 41) provide in-depth discussions of the theoretical and technical elements associated with the filtration of fluids with the goal of removing microbial and viral contamination. EMA Annexes 1 and PIC/S Annex 2A state compliance expectations when the drug product cannot be filter-sterilized, as in the case in cell therapies. Where filtration is not possible, aseptic processing is expected. In addition, the sterile filtration or aseptic processing of starting materials is also expected (e.g., media used for producer cells, transfection agents, other additives). The decision to aseptically produce without filtration must be accompanied by a risk assessment, including results of the filter development study from the specific product or from platform experience, that demonstrate the product was irreparably harmed by the process of filtration. For more detailed information, see FDA *Guidance for Industry: Sterile Drug Products Produced by Aseptic Processing — Current Good Manufacturing Practice* and EU Annex 1.

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Q 4.2. What are the key considerations for the control strategy when filtration is considered and when filtration cannot be performed?

The following are key elements to deliberate when filtration is considered:

- Intended purpose of the filtration step (e.g., sterilization, bioburden reduction, viral removal)
- Type of filter and its material of construction (e.g., depth filter, casted membrane, hollow fiber, cartridge)
- Process conditions (e.g., temperature, pressure differential, flow rate, nature of the fluid, flush volume)
- Nature and level of microbial and/or viral contamination
- Quality impact on the product (e.g., yield, adsorption, viability/activity, inactivation, matrix reactivity, aggregates, extractables/leachables)

These elements must be evaluated as part of the control strategy when filtration is identified as the mechanism to remove adventitious agents (e.g., microbial and viral contamination). These elements should also be considered as part of the demonstration that filtration was not possible if a risk assessment is planned for that purpose. (See PDA TR 26 for additional information.)

Whether the plasmid or vector will be filtered or aseptically manipulated without filtration, the control strategy is expected to minimize opportunities for adventitious agent contamination. To ensure that the control strategy is sufficiently robust, a thorough evaluation of the process from feed to final product is required to ensure the proper controls are in place to prevent adventitious agents from being introduced into the product stream. Each step should be evaluated to identify aseptic manipulations and specify the requirements for either reduced bioburden or sterile/virus-free components. In early development laboratories, aseptic processing may seem to be impractical, even though GLP and/or GxP quality systems are in place. If vector is produced in an R&D, academic, or process-development lab and later used in first-in-human trials, however, careful consideration through a risk assessment and an appropriate quality assurance review is warranted to evaluate the potential for microbial and viral contamination and patient risks. PIC/S Annex 2A provides guidance on the design and separation of ventilation systems that should be considered when multiple replication-competent vectors are being developed in parallel with other processes.

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Q 4.3. What is the impact of sterile filtration on plasmids and vectors?

Removal of microbial contamination is commonly attained by means of 0.1 µm to 0.22 µm membranes. The structure of plasmids and vectors make them susceptible to aggregation during certain processing conditions. Close attention must be paid to steps prior to the filtration that could result in the formation of aggregates as the solutions approach the membrane. If aggregates are formed, they may be retained on the filter resulting in reduced yield and potency.

Working with membrane vendors during development may enable the reduction of harm to the product (e.g., preventing product loss, the formation of aggregates, loss of potency). Fine-tuning the product chemistry and the membrane characteristics may yield improved results. For example, with lentiviral vectors, performing scaled-down studies with 43 mm disc membranes from several different vendors may be beneficial to study the differences in impact that different membranes impart. The use of scaled-down studies economizes on the amount of vector used during development when it is in short supply. Often dual-layer, asymmetric polyethersulfone membranes are successful candidates for developing a lentiviral control strategy which includes final filtration.

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Q 4.4. Why is the use of nanofiltration to mitigate viral contamination relevant?

Virus-retentive are designed to remove potential viruses by means of size exclusion; the nominal pore size in these filters is 0.1 µm. These filters are used in ATMPs to process culture media or buffers used to culture or expand cells where viral contamination will be detrimental to the product, and they should be considered as part of the overall adventitious virus control strategy.

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Q 4.5. Does the sterile filter have to be the last step in the process before filling?

Ideally, the point of filtration should come after all significant additions and manipulations of the product, which naturally places the filtration step as the last activity prior to filling. If a different position within the process is warranted (e.g., based on harm to the product or excessive product loss during filtration), a risk assessment may be possible to support an earlier position for filtration in the process. The final position for the filtration and its justification should be part of the formal control strategy.

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Q 4.6. In early vector development, what is the timing for the decision on whether to aseptically process or sterile filter?

The decision to filter or to aseptically process should be made as early as practicable in the design of the product. The earlier the decision is made, the easier it will be to ensure that the appropriate control elements are included in the CCS and that the necessary unit operations are protected as required to fulfill the design requirements. However, the decision to not include a sterile filtration step requires justification, appropriate risk assessment, and data to support the risk assessment, therefore, the timing may depend on the collection of data and justification of the approach. Ultimately, the final approach should be codified within the QTPP, the control strategy for the process, the microbiological and virological CCS, and the facility controls.

Early vector development commonly occurs in shared facilities that may have more than one standard for processing: therefore, careful procedural controls should be in place for each material developed that reflect the appropriate use of those materials (e.g., animal studies, tox studies).

Different regulatory authorities have varying stances on sterilization process validation and the timing relative to the development of clinical materials. For instance, in Europe there is an expectation that sterilization validation will be complete and available prior to Phase I (see **Q 4.1**). The regulatory requirements for the jurisdictions selected by the MAH may therefore influence the timing of the decision on aseptic processing or filtration.

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Q 4.7. When sterile filtration is not an option, at what point in the process is aseptic processing needed?

In an aseptic process, the materials used in the process are sterilized separately (e.g., filtration, heat, irradiation) and then brought together in an aseptic manner. The point where aseptic manipulations must be conducted should be as far downstream in the process as practicable. In this manner, there is a high degree of sterility assurance for the individual components and the aseptic processing steps (the aseptic boundary) are kept to the absolute minimum.

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Topic 5. Comparability

Q 5.1. How are bridging studies different from comparability studies?

Bridging and comparability are terms commonly used during the development and registration of a new product when comparing and assessing differences between processes or materials. They are used separately (e.g., bridging of reference standards, comparability of lots from different sites) and, more confusingly, can also be used together such as in *ICH Quality Guideline Q5E: Comparability of Biotechnological/Biological Products Subject to Changes in their Manufacturing Process* where there is a reference to a "comparability bridging study" ***(14)***. Therefore, it is important to first define these terms in order to offer clarity about their objectives and execution.

Bridging is the process of relating earlier data to later data in a way that the ability to extrapolate earlier results to later results is maintained. Common usage includes bridging of clinical studies, bridging of reference standards, or bridging from one test method to a new version of that test method.

Comparability is the process of comparing technical data from a reference population to a new population for the purpose of ensuring equivalence in quality, efficacy, and/or safety. Common usage includes comparability of product manufactured by different manufacturing sites, particularly when launching at a new site. It may also include comparability (comparing) product quality attributes and patient safety data after a significant process change.

In many situations, bridging is achieved when analytical comparability of product quality or analytical method outputs before and after a change is demonstrated. When comparability is not demonstrated by only analytical testing, additional nonclinical and clinical bridging studies are needed.

When the terms are used together in ICH Q5E, comparability bridging studies include aspects of both of these definitions. "A study performed **to provide nonclinical or clinical data** that allows extrapolation of the existing data from the drug product produced by the current process to the drug product from the changed process [emphasis added]." In this case, the intent is to be able to use the results of equivalence to enable the extrapolation of prior results. From the perspective of regulatory affairs and regulatory bodies, comparability in this sense also may represent the formal agreement on reduced reporting requirements for a manufacturing site change, for example.

In early phases of development, establishing in-house or interim reference standards that are useful for the development work, but are not intended for commercial use, is common. When establishing a primary reference standard and subsequent working reference standard, conducting bridging studies of the reference standards to be able to relate lot test results from early safety and clinical work to the testing from batches that represent the commercial process is essential. This can become critical for materials that are stored refrigerated rather than frozen as the stability of older samples may not be maintained in a way so they can be retested at a later date.

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Similarly, test methods might evolve throughout the development and commercialization program as they are refined or improved (e.g., enhanced method controls or automation, improved precision, greater limits of quantification or detection). Bridging of methods ensures that results from earlier generations of test methods can be interpreted in the context of results generated with the new versions of the test methods.

Nowhere is bridging more critical than for stability studies. Data generated with reference standards or test methods at earlier timepoints must be preserved through an understanding of the bridging to the current methods and, therefore, the results at the current and future timepoints. Often stability studies can be performed with both the old and new methods (and old and new reference standards) in parallel from the point of introduction of the new method through the endpoint of the study. This will reinforce the conclusions from the bridging study performed and provide a better understanding of the stability conclusions overall.

Comparability is performed to assess the quality, efficacy, and/or safety attributes of a lot(s) against other lot(s) that are differentiated by process, site of manufacture, raw material, or other potentially significant difference. To perform comparability, extensive analytical characterization data are collected and compared to establish equivalence. At early stages of development, comparability may be required due to differences, for example, between Phase I and Phase II clinical material and the intended Phase III process. These comparisons of analytical consistency may be made more complicated by changes made to the methods or reference standards themselves, which highlights the importance of method and/or reference standard bridging studies.

Some important points to consider for bridging and comparability include the following:

- Ensure that bridges and comparability are always maintained to clinical materials, especially the pivotal lots that will demonstrate efficacy, safety, and/or immunogenicity
- Keep careful track of which lots were tested with which version of the assay or reference standard and know what those differences might mean in terms of bias or offset
- Establish written guidelines for when CoAs will or will not be updated based on the introduction of new test methods or new reference standards, and ensure that there is a clear signifier on the CoA that traces to its version that permits ready traceability to the testing that was used to generate the values
- Ensure that with each intended change there is a planned strategy for bridging stability data to new methods and reference standards to preserve the integrity of earlier timepoints, before the change is implemented
- Use bridging and comparability protocols to define the criteria by which comparisons and conclusions are to be made
- Ensure that an inventory of clinical lots, reference standard lots, and other representative lots are set aside and maintained for the purpose of establishing the essential bridges required for the program and, for every designated use of these limited supplies, ensure that there is an adequate protocol and justification for their intended use

Establishing standard operating procedures that include these requirements is one way to ensure success in demonstrating that product, processes, reference standards, and/or analytical results are consistent and traceable.

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Q 5.2. How can a change control strategy compare or utilize test results from two sites, from two batches, or from clinical material to process performance qualification material?

Demonstration of comparability of product quality attributes and patient safety data from product manufactured under different conditions (e.g., different development stages, different site of manufacture, different analytical methods) is critical to prove that previous preclinical toxicology, nonclinical, and clinical study results are still relevant. Further, demonstration of comparability will prove that the same control strategy, specifications, and shelf life can still apply to the post-change products. Therefore, ICH Q5E is also applicable to ATMPs.

For analytical comparability, typically three levels are involved. The first level is lot-release testing, where comparability criteria should be established based on historical data generated from materials manufactured before the change. In many cases, the comparability criteria might need to be tighter than the release specifications. To meet the predefined comparability criteria, not only should all lots made from the new process meet release specifications, but they also should be within the comparability criteria.

The second level is characterization testing, wherein additional characterization assays should be performed in addition to the lot-release testing to demonstrate there are no other quality attribute changes that are not detected by lot-release testing. A significant change in any quality attribute (critical or noncritical) may suggest there may be other underlying changes that could pose risks to product safety and efficacy.

The third level is stability comparability, which could include stability studies at recommended storage conditions and/or accelerated/stress stability studies. Comparability of stability (both degradation rate and profiles) should justify that the previously established shelf life can still be applied to the post-change product. In addition, accelerated/stress stability studies are considered a “pressure test” to reveal any differences that may not be detected under long-term storage conditions during routine analysis. Such stress/forced degradation studies are commonly done for therapeutic protein products during a comparability assessment. Critical quality attributes, degradation pathways, and impurity profiles of many ATMP products remain to be fully understood and elucidated, so the use of these studies in comparability have not been widely published.

The stringency of the comparability exercise increases as the product progresses into later stages of development. Comparability of early preclinical/toxicology to first clinical materials enables the first-in-human clinical trial, while comparability between pivotal clinical materials and validation (process performance qualification (PPQ)) is essential to a smooth Biologics License Application (BLA) review and approval process. It should be recognized, however, that even the strongest comparability evaluation does not preclude the risk that regulators will request an in vivo assessment at later stages of product and process development.

Many ATMPs are for rare disease indications, so the number of lots manufactured at early stages may be limited and it could be difficult to establish meaningful comparability acceptance criteria with such a limited data set. Increasing the number of replicate tests for the same material could help set criteria that take assay variability into consideration, which is especially important for critical, yet highly variable, biological assays (e.g., titer and potency). Establishing assay controls in early stages is crucial to understand, monitor, and control the assay performance, particularly when the assay is run in different laboratories or sites and when results are to be used in the comparability exercise. Method transfer protocols and acceptance criteria need to be designed carefully to ensure the sending unit and receiving unit obtain equivalent results while still

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allowing for normal method variability. Method validation will serve as an essential yardstick for method performance at all sites, therefore, it is essential that the method validation represents the true capability of the assay precision, intermediate precision, and accuracy. Analytical methods used in different laboratories (e.g., different CROs) can be assessed retrospectively using assay controls, reference standards, and the same sample set provided sufficient accommodation has been made to preserve benchmark lots for this purpose.

Unique challenges exist for ATMP comparability studies. The quantity of ATMP samples is typically limited and may not be sufficient for a comprehensive comparability testing plan. However, new analytical technologies (e.g., charge detection mass spectrometry, small-volume subvisible particle method) that consume only small sample volumes are advantageous as their use will minimize the amount of product used from the manufactured lot.

As the focus of the process/product comparability exercise is to demonstrate product comparability, certain practices can be adopted during analytical testing:

1. Test pre- and post-change products side-by-side within one laboratory and within one run/plate/sequence, if possible, to minimize assay variability. Assay setup can be adjusted to accommodate such needs (e.g., splitting two samples equally between two plates instead of using one plate for each sample).
2. Use reagents (especially critical reagents) from the same vendor or even use the same lots where possible.
3. When side-by-side testing is not possible due to material scarcity and product stability, it is beneficial to include a historical sample(s) (e.g., reference standard or assay control) in the testing of post-change products to monitor potential assay drift and justify the apparent difference it causes.

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Q 5.3. What approaches can be used when it is necessary to change test methods during a comparability protocol?

Changing test methods during product development is common. This is especially true for ATMPs as these therapies are relatively new and the analytical methods and associated controls necessary to fully monitor and assure product quality are still lagging behind other biological products, such as monoclonal antibodies.

Test methods should be defined and understood prior to assessing comparability. Therefore, it is undesirable to change both the methods and the attribute to be studied for comparability (e.g., site of manufacture); however, it is understood that ongoing development may necessitate changing or updating test methods at times. When updates are necessary, method bridging can be used as one approach to understand changes in test methods. Method bridging studies can be accomplished by testing the same set of samples by both methods. The method bridging is important to demonstrate method comparability. Including all relevant sample types (especially the stability sample) is essential to ensure thorough understanding of the method differences. While less than ideal, it is acceptable to have bias between methods as long as the impact of the bias to the control strategy—in-process controls, release and stability specifications, and comparability criteria—is adequately assessed. Well-characterized method bias should be factored in when setting up comparability acceptance criteria if historical data generated using the old method will be used to compare against new data generated with the new method.

Both process comparability exercises and method bridging studies highlight the importance of keeping retention samples. This is extremely challenging for ATMPs since, during early development, the limited material available is needed for various purposes by multiple teams (e.g., analytical development, reference standard, stability, preclinical studies). For some of the assays, a platform approach or a product independent sample/control can be adopted. For a host protein or a DNA assay, for example, a sample from a different product using the same or similar manufacturing platform process could be used in the method bridging study. Sample stability after long-term storage needs to be taken into consideration when side-by-side retesting is performed on historical samples together with recently generated samples. Differences may arise from the stability storage of the historical sample retained. In such situations, historical results or profiles at, or close to, time-zero samples should be leveraged to demonstrate comparability.

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Topic 6. Use of Platform Technologies for ATMP Products

Q 6.1. How can platform technologies be used to improve quality and efficiency in the development of ATMP products?

A platform is a model process (or model analytics) from which variations are developed. The term "platform" was adopted as the model represents the jumping-off point for any new development efforts. For example, a model process would have identical unit operations for the majority of the process steps with the same process controls (the process backbone) with only targeted changes, such as an adjustment to an individual process parameter or setpoint, to adapt to the needed conditions specific to the biological entity (e.g., strain).

Within the industry, the term platform may also be applied to specific unit operation equipment technology. Although used in this context, the terminology may be imprecise in some cases, as the use of a particular piece of equipment or group of equipment does not ensure that the same process or analytics are maintained. While these equipment "platforms" can help to standardize and streamline qualification, validation, and overall control strategies, they may not provide the same level of regulatory efficiency that are being described in this section.

The development of multiple products based on one or more common technologies, or platforms, offers an approach to effectively use prior knowledge from existing products as new products are developed. The interest in pursuing platform technologies has risen significantly in the last several years, particularly for ATMPs, as these products are typically developed under aggressive timelines due to the nature and promise of the therapeutic effect. The use of platforms can allow for:

- Standardization during manufacturing process development
- Standardization during development of analytical methods
- Establishment of harmonized CQAs and the associated risk-based assessments
- Establishment of consistent process and analytical control strategies
- Harmonization of regulatory submission content and technical support documents
- Reduced costs and timelines associated with development activities
- Reduced time for review and approval by regulators (when prior art has already been approved)
- Efficient process qualification approaches (e.g., the inserted coding region in a common viral-vector or plasmid drug substance is produced and purified using the same overall production and control processes)

The U.S. government has pilot programs promoting the development of platform technology (e.g., the "Paving the Way for Rare Disease Gene Therapies" project) as it is understood that the benefits to patients from the therapies, and the possible shortening of the time to market (which for traditional development pathways has been more than 7–10 years), is in the best interest of the public ***(15)***.

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The use of a technology platform also contributes to enhanced process knowledge on an ongoing basis as data from additional molecules become available and can be used to further characterize and refine a platform.

The use of platforms is consistent with QbD (see **Topic 2**) and robust control strategies (see **Topic 3**) as there is a heavy emphasis on prior knowledge and a reliance on effective knowledge management to collect and organize the critical data that ensures appropriate conclusions can be drawn to strengthen the knowledge of the design space (i.e., the combination of critical material attributes and critical process parameter ranges that define the successful product manufacture) and the performance of the process.

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Q 6.2. What types of ATMPs are most likely to benefit from the application of platform technologies?

For different ATMPs, the decision to use one or more platforms is driven by different factors including stage of development (proof of concept versus licensed product), cost, target disease, and intellectual property.

ATMPs are a highly diverse group of products and, consequently, there is a range in the extent to which platform technologies can be applied. AAV vectors are an example of a product class that makes extensive use of platform technologies, where the manufacture of an AAV vector may use the same production methods and controls as the manufacture of a different product with the same vector serotype and different gene-of-interest sequence. Different technologies for manufacturing AAVs have been used to develop a range of products, making it possible to use this prior knowledge to develop an understanding of how the process needs to be controlled. In contrast, an ex-vivo genetically modified cell product relies upon lot-specific starting materials and a number of different technologies are used to create a wide range of custom product types that are classified as ATMPs.

For pDNA used as a drug substance, FDA and EMA have accepted efficient PPQ approaches when multiple pDNA expressing different antigens are a) produced from the same backbone plasmid vector, b) produced with identical unit operations with the same master batch-record and testing, and c) are combined in a single drug product. In such examples, regulatory agencies have agreed to fewer than three consecutive runs for each plasmid individually, based on a matrix approach for PPQ of the platform. In another example, in the vaccine industry, the production of drug substances with different serotypes or strains are not independently validated, but specific parameter targets are adjusted and confirmed via initial production batches.

Similarly, for CAR-T products/lentivirus platforms, four or more different vectors that share a common backbone and yet differ in the cell-surface-receptor-coding region and potency testing, have been filed in an IND as a platform using QbD principles, avoiding repetitive CMC sections in each BLA.

Platform technologies are used extensively both in the production and testing of ATMPs. For some products, individual manufacturing or testing activities may be based on the platform and, in other cases, platform technologies may account for the majority of all activities. For AAV vectors, for example, the platforms may differ based on such elements as cell lines, AAV serotypes, and/or construction methods.

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Q 6.3. What are the considerations for developing and using platforms for ATMP manufacturing?

Any platform-leveraging approach should make use of strong risk-based rationales for all variations in processes and process control. The magnitude of the risk should drive decisions regarding when supporting studies may be required.

There is a need to carefully evaluate all changes from the baseline platform during development to determine which foundational studies may need to be repeated, in whole or in part, due to platform changes. For example:

- Changes to upstream processing may result in the need to recharacterize cell-expansion curves, feed regimes, end of production cells, or limit of in-vitro cell age
- Changes to downstream processing or the purification process parameters might require reevaluation of baseline studies such as small-scale viral clearance and small-scale resin lifetime
- Changes at any point in the process may result in an incremental change to the analytical methods, so it is important to ensure that appropriate bridging and comparability are performed (see **Topic 5**)

While a platform may be able to be filed as a common control strategy, it is important to remember that individual products will require separate BLAs related to the individual product's clinical-trial results and control-strategy-specific variations.

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Topic 7. Potency

Q 7.1. What are some of the key attributes of an effective product-specific potency assay?

Potency is an expression of the activity of a drug substance or drug product in terms of the amount or concentration required to produce a defined effect. Clinical efficacy, on the other hand, is a measure of the therapeutic effectiveness of the drug in humans. Regulators desire a product-specific potency assay that provides a demonstrable link between potency and clinical efficacy. As such, the development team should first identify the relevant biological property linked to the product's in vivo mechanism of action (MOA) and then establish a method to provide a quantitative measure of that biological property. The assay must demonstrate that its performance remains unaffected by expected variability in product purity or matrix effects.

Other than safety, potency is the most critical quality attribute of a product. The ideal potency assay should reflect the MOA of the drug and demonstrate the product has the expected ultimate “functionality” that correlates with clinical efficacy. For an in-vivo gene therapy product, the ultimate functionality means the action of the expressed (or transcribed) gene of interest (GOI). If the product is a protein, then demonstration of the expected bioactivity (e.g., enzymatic activity) of the expressed protein is the ultimate goal of potency assay development. For an activated cell therapy product, for example, this broadly means that there must be an in vitro analytical method that correlates to in vivo reduction of tumor size or number or reducing the progression to metastasis.

Potency is a CQA for stability, not just release; therefore, an effective potency assay must also demonstrate that it is stability-indicating. To do this, forced degradation studies should be performed which demonstrate the primary degradation pathways of the product. With the samples showing between 10% and 20% degradation for each condition (where possible), as confirmed by orthogonal methods, the potency assay must be able to accurately detect the associated loss in potency.

If the product will be produced across multiple sites, it should be recognized that small differences in process and materials may exist site to site. Ideally, the potency assay should be robust enough to account for these small differences when they do not represent a difference in clinical effect.

To effectively quantitate potency, it is necessary to establish a reference standard against which the response from each lot will be quantitated. Initially, the reference standard may be a well-characterized development lot as an interim reference standard. However, as product development advances, the company must qualify a primary reference standard that will be used for commercial production. Bridging will likely be required to link the quantitation of early lots (and their stability studies) to any new reference standard that is introduced (see **Topic 5**). It is especially important to maintain a quantitative reference to clinical lots.

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Since development of an appropriate potency assay is complex, it is recommended to collect a significant amount of bioanalytical characterization data on early product lots until it is demonstrated that the developed potency assay is robust to the product and process and is predictive of MOA. In this manner, should the potency assay undergo changes later in development, the available bioanalytical data can help support the bridging.

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Q 7.2. What are acceptable approaches for phasing in a product-specific potency assay during development?

Due to the complex nature of ATMP products, developing a potency assay that is truly reflective of the MOA of the product is challenging, especially in the early development stage. Some of the GOIs encode structural proteins that lack clearly indicated bioactivity, while the functions of other GOIs are multifaceted, while some are not yet fully understood. Regulatory agencies are aware of such challenges and typically allow for flexibility in determining the most appropriate potency assay for each product at each development stage. The expectations for the potency assay increase proportionally as the drug development progresses.

A widely accepted approach for potency development is the matrix approach. A combination of various assays is used to assess product potency holistically. It starts with the measurement of a DNA copy number, vector-particle number, cell counts, and cell-viability physical attributes that are necessary (but may not be sufficient) for potency. As product development progresses, the matrix evolves, and certain methods may drop out due to complexity or robustness concerns. Other assays developed in parallel that are more reflective of MOA and demonstrate functionality of the product may become more central to the testing scheme. The matrix can comprise both quantitative and qualitative assays. For example, a binding assay can provide quantitative results compared to the reference standard while a cell-based assay can confirm the expected functionality (pass or fail). The correlation between the quantitative and qualitative assays can be established during assay development to justify the use of the combination of quantitative and qualitative assays in routine release.

In specific cases where there is no ability to directly measure MOA, it may be possible to use more than one assay to represent potency, providing regulatory agreement is granted. For example, in the specific case of pDNA or mRNA as the drug product, there is no specific in vitro assay indicating MOA because transcription and translation of the drug product for presentation to the immune system is required in vivo. In these cases, FDA has accepted direct measurement of nucleic acid coupled with integrity of the molecule (e.g., supercoil measurement) as surrogates for potency assay.

The ideal potency assay, especially one for routine quality control release, needs to balance its capability and reliability. A complicated, research-grade assay that can only be conducted on a sophisticated instrument by a highly trained scientist is not feasible for routine quality control testing of the product. A simple, robust, and reproducible assay is preferred. A robust data package is expected to justify using a surrogate assay for functionality potency assay. A correlation study using a regular drug substance or drug product, an in-process degraded sample, and/or specially constructed products with reduced or elevated potency would be valuable to support the assay choice. Open and timely communication with regulatory agencies on a potency assay strategy is of critical importance for the success of a project. Many clinical trial applications and licensure approvals have been delayed due to potency assay concerns in recent years. Recently, FDA proposed a multifaceted approach combining manufacturing process design, process control, material control, in-process testing, and potency lot release assays as a systematic potency assurance strategy ***(16)***. Sponsors need to prepare for a holistic overview of potency assurance in addition to a robust, MOA reflective potency assay.

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Points to Consider No. 11: Development, Classification, Manufacture, Control, and Testing of Plasmids and Vectors Used in ATMP Production

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